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**National Social Life, Health, and
Aging Project (NSHAP): Wave 1,
[United States], July 2005-March
2006**

Technical Report: Vaginal Swabs

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National Social Life, Health, and Aging Project (NSHAP): Wave 1, [United States], July 2005-March 2006

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Bacterial Vaginosis

Vaginal Swab Measurement of Bacterial Vaginosis in Wave I of the National Social Life Health & Aging Project

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Rationale

Bacterial Vaginosis (BV) is a common polymicrobial condition of the vagina that occurs with reduction of vaginal acidity (pH) and a shift in the normal bacterial flora. In the younger population, BV is associated with douching (LlahaCamp, Rai et al. 1996), sexual intercourse with multiple partners, oral sex with different sexual partners, and cigarette smoking (Smart, Singal et al. 2004). Furthermore, BV in younger women has been associated with infection in the lining of the uterus and upper genital tract, which in turn, can elevate the risk of infectious complications following the gynecologic disorder (Lindau, Hoffmann et

al. Under Review). BV also increases the risk of sexually transmitted infections such as HIV (Sewankambo 1997; Sobel 2007). While BV is common among women of reproductive-age, information is limited regarding BV prevalence in older women (Allsworth and Peipert 2007).

Measurement

Clinical diagnosis of BV can be characterized by the following criteria: 1) thin, off-white vaginal discharge with a “fishy” odor documented by amine testing 2), abnormal cervical discharge and vaginal bleeding (especially after intercourse) 3) a pH in excess of 4.7, and 4) clue cells on microscopic evaluation of saline wet preparation (Sobel 2007 Lancet). These criteria are based on Amsel’s work, and are still used today (Amsel, Totten et al. 1983; Gutman, Peipert et al. 2005). Alternatively, Gram staining is performed on a vaginal smear and scored using either the Hay/Ison (Ison and Hay 2002) or Nugent criteria (Nugent, Krohn et al. 1991).

Population Prevalence

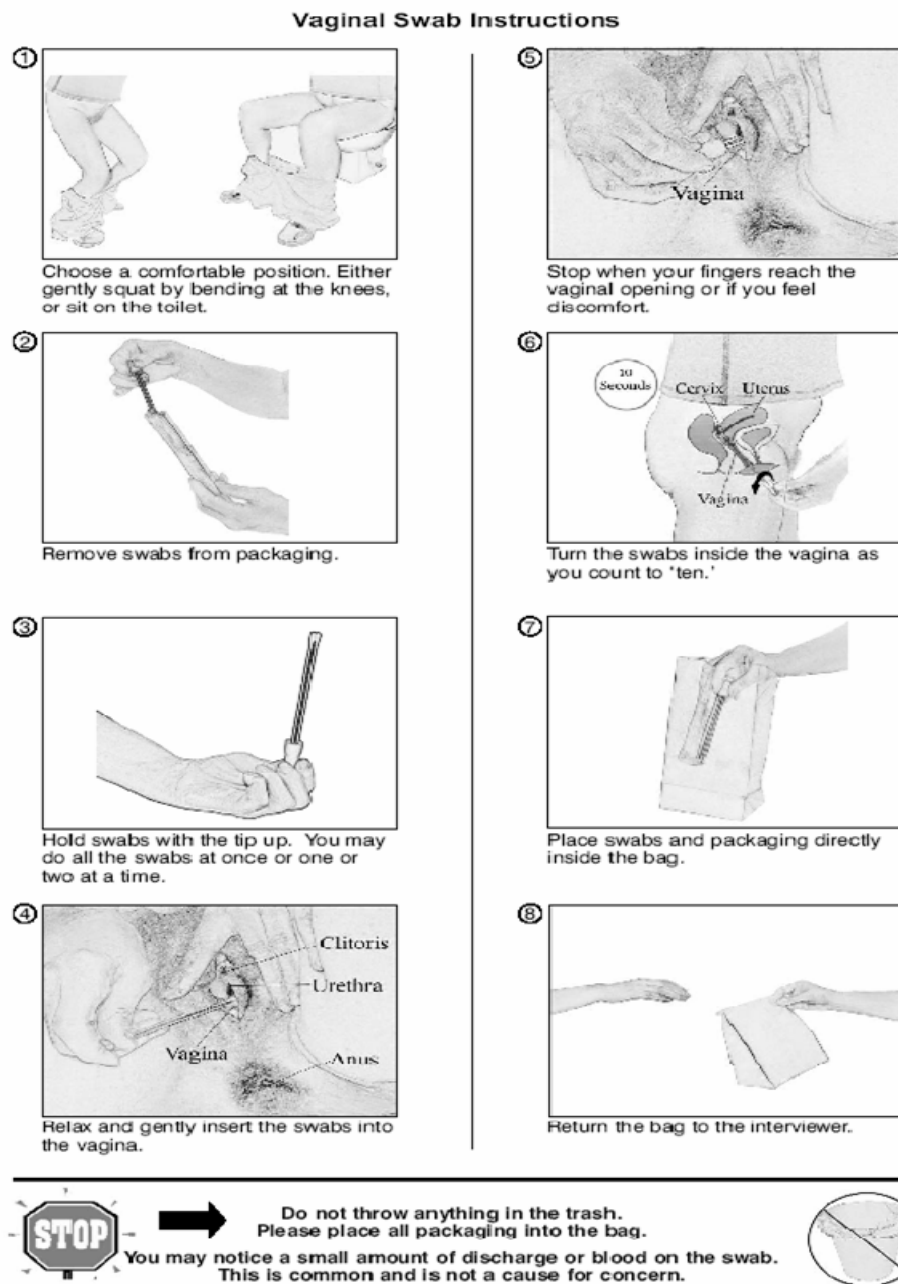
Bacterial Vaginosis is a common vaginal disorder among women (Sobel 2005). It accounts for up to 50% of cases of vaginal discharge in non-pregnant women. (Sobel, Ferris et al. 2006). However, BV may be asymptomatic (Sobel 2007 Lancet) or variably symptomatic (Klebanoff, Schwebke et al. 2004).

In the data from the National Health and Nutrition Examination Survey, BV prevalence was estimated to be 29% among the general population of women ages 14-49, using a self-swab protocol. The prevalence among younger women (14-19 years) was lower (23.3%) than among women 40-49 years (31.3%) (Allsworth and Peipert 2007). Data regarding BV prevalence in post-menopausal women are limited (Cauci, Driussi et al. 2002). In a northern Italian study among non-pregnant women (40-79 years) 6% of postmenopausal women, 11% of perimenopausal women, and 9.8 % of fertile women met microbiologic criteria for a diagnosis of BV (Cauci, Driussi et al. 2002).

Specimen Collection (Females Only)

The Vaginal Swab Protocol was designed to collect vaginal specimens for Bacterial Vaginosis (BV), Vaginal Candidiasis (VC), Human Papillomavirus (HPV) testing, and vaginal cytology analysis. All female respondents were asked to provide a vaginal self-swab specimen. Procedures were explained using a scripted description aided by illustrated instructions, developed for NSHAP by a medical illustrator in conjunction with a study investigator who is a gynecologist (Figure 1). Participant questions were addressed using a “frequently asked questions” document to ensure consistency of responses across field staff. Field staff read each step of the illustrated instructions to the respondent and asked for questions. Participants were given the instruction card with the collection materials (Female Swab Specimen Collection Kit, Catalog No. 5123-1220; Digene Corporation, Gaithersburg, MD and BBL™ CultureSwab™ Plus, Catalog No. 220117; Becton, Dickinson and Company, Franklin Lakes, NJ) and directed to a bathroom or other private room in the home. When the respondent returned, the interviewer then secured the Digene swab inside a tube containing 1 mL Specimen Transport Medium™ (STM; Digene Corp.) and the BBL™ CultureSwab™ inside a tube containing Amies medium without charcoal (Becton, Dickinson and Company). The interviewer labeled both tubes with the unique, numeric identification number. At the end of each home encounter, field staff stored the vaginal swab transport tubes in an insulated cooler with ice packs. Vaginal swabs were shipped daily on cold packs in a Styrofoam container to the University of Pittsburgh, Magee-Women’s Hospital Department of Pathology clinical microbiology laboratory via overnight delivery. The swabs were packaged in accordance with the federal shipping guidelines for diagnostic biological material. Following processing at Magee-Women’s, one BBL™ CultureSwab™ for each respondent was repackaged and shipped overnight on cold packs to the University of Chicago Institute for Mind and Biology laboratory for cytological analysis. An interactive reconciliation system facilitated remote tracking of vaginal swabs. Vaginal swab specimens were collected from all willing female respondents (n = 1,028), with an adjusted cooperation rate of 67.6% (Lindau, Hoffmann et al. Under Review).

Figure 1. Self Collection of Vaginal Epithelial cells using Dacron Swab



Instructions developed by medical illustrator Rachel Seelen in conjunction with Stacy Lindau, MD.

Shipping and Storage

After collection, the vaginal swab tubes were given to the interviewer and stored at 2-8°F, in an insulated bag with two reusable ice packs.

Prior to shipping, collection tubes containing specimens were removed from the insulated bag, placed in a small Ziploc bag with a handful of cotton balls, and then placed in a Styrofoam™ container and 8" x 7" x 7" cardboard box with a disposable ice pack. They were shipped to The University of Pittsburgh Department of Pathology, Magee-Women's Hospital Clinical Microbiology Laboratory by FedEx Express, by placing the package in a drop box or calling for pickup by FedEx. Specimens were shipped daily.

<i>Method</i>	FedEx Express: placed in drop box or picked up by FedEx
<i>Shipping Address</i>	Jeanne Jordan Magee-Women's Hospital Clinical Microbiology Lab 300 Halket Street, Room 4680 Pittsburgh, PA 15213 (412) 641-4104

Assay

A Gram stained slide of a smear made from the vaginal swab samples was made in the Magee-Women's Research Institute laboratory of JAJ. The smear was evaluated for bacterial morphotypes and interpreted in accordance with pre-existing criteria. Nugent's criteria (Nugent, Krohn et al. 1991) for a standardized 0-10 point scoring system were used to interpret the Gram stain result. Zero to 3 was classified as normal/no BV detected, 4-6 as intermediate, and 7-10 indicated the presence of BV. Each stained slide was read by two independent observers who were blinded to the other's results. Discordant results were re-reviewed by the initial two readers as well as reviewed by a third observer to resolve the discrepancies.

Table 1. NSHAP Vaginal Swab testing performed by Magee-Women's Research Institute			
Assay Type	Assay principle	Regulatory Status	Samples
Gram stain smear	Evaluation of bacterial morphotypes	None	Vaginal or cervical self-collected swab

The same gram stained slides used to assess the presence of bacterial vaginosis were also evaluated for the presence of yeast, by using the 1000x objective. A dichotomous score was assigned to each specimen slide, after the reader viewed and examined the entire area of each stained slide to determine the presence or absence of yeast cells showing blastoconidia (cell buds)(Lindau, Hoffmann et al. Under Review).

Technical details of the vaginal swab laboratory assays have been previously described (Lindau, Drum et al. Under Review). The BV assay technique is summarized in the following table

Table 2. Summary of Vaginal Swab Assays Obtained from Self-Swab Specimens in NSHAP[†]					
	Collection Device	Manufacturer	Laboratory	Assay Type	Protocol
Bacterial Vaginosis (BV)	Double Copan swab with Dacron tip	BBL™ CultureSwab™ Plus, Catalog No. 220117; Becton,	UP-MWRI	Gram stain of vaginal material on glass slide (Burke, 1922)	Clinical

		Dickinson and Company, Franklin Lakes, NJ			
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Quality Control

The laboratory participates in semi-annual proficiency testing for gram staining and BV interpretation, and must achieve greater than 90% overall agreement with these challenges. Gram stain reagents are checked weekly and with each new lot of stain before they are put into use. Gram stain reagents were evaluated by staining ATCC 25923 Staphylococcus aureus, ATCC 25922 E. coli, and ATCC 26555 Candida albicans.

Availability

	STM Vaginal Swab	Blue-Tipped Vaginal Swab (double swab)
Product Name	Female Swab Specimen Collection Kit *Collection kit includes: sterile Dacron® swab and tube with 1 mL of Specimen Transport Medium™	BD BBL™ CultureSwab™ Plus Amies without Charcoal
Manufacturer	Digene Corporation	Becton, Dickinson and Company
Location of Manufacturer	Gaithersburg, MD	Franklin Lakes, NJ
Product Number	5123-1220	220117
Interviewer Instructions	When respondent returns, insert the swab into the STM tube and break off the extra handle of the swab by pressing it along the side of the tube. Label the tube with Respondent's SUID number using a Sharpie pen and a blank lab label.	When respondent returns, remove cap from tube with gel and tightly insert blue tipped swab in gel. Label tube with Respondent's SUID number using a Sharpie pen and a blank lab label.

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Human Papillomavirus (HPV)

Vaginal Swab Measurement of Human Papillomavirus in Wave I of the Social Life Health & Aging Project

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Rationale

Human papillomavirus (HPV), a common sexually transmitted viral infection of the vagina and cervix, is an important factor in genital tract dysplasia (precancer) and is found in association with nearly all cases of cervical cancer (Schiffman, Bauer et al. 1993). More than 100 types of HPV have been identified, with as many as 40 infecting the female genital tract (Munoz, Bosch et al. 2003). HPV infection in younger women is frequently transient but can be persistent, latent, or reactivated. Risk factors for genital tract HPV in women include sexual activity, tobacco use, and immune suppression (Bosch, Lorincz et al. 2002). NSHAP provides data on the presence and subtypes of high risk HPV, or the strains that are most strongly associated with cervical dysplasia and cancer. Genital tract HPV presence may be a marker of immune function in older women (Garcia-Pineres, Hildesheim et al. 2006).

Measurement

High-risk (HR) HPV DNA testing was performed using the HC2 assay method (Digene Corp.) and rapid capture system (RCS) according to the manufacturer's protocol. The HR HC2 signal amplification assay contains a cocktail of probes complementary to 13 HR HPV types including 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59 and 68. In addition to the kit controls, an external sample processing control consisting of approximately 5,000 HeLa cells, which contain HPV 18 DNA was processed with each batched of specimens to control of adequate cell lysis. Specimens with Relative Light Unit/Cutoff Values equal to or greater than 1.0, obtained using the DML 2000 instrument were considered positive for HR-HPV DNA. (Lindau and Drum Under Review)

Population Prevalence

No population-based estimates are available for older women in the U.S.; however, clinical studies of post-menopausal women suggest that 10-20% have any genital tract HPV infection. Research outside the U.S. suggests that prevalence of high-risk subtypes is lower, ranging from 5.8% cumulative frequency of HPV-HR positivity among women 55 and older (Smith, Ritchie et al. 2003) to 13.6% for oncogenic subtypes for women age 65+ (Herrero, Castle et al. 2005)

Table 1: Human Papillomavirus Prevalence in Older Women (%)		
Rates of HPV Positivity*		
Age	Smith et al. (2004)	Smith et al. (2003)
<55	13.0	13.5
55-59	20.3	15.7
60-65		12.4
65+	N/A	

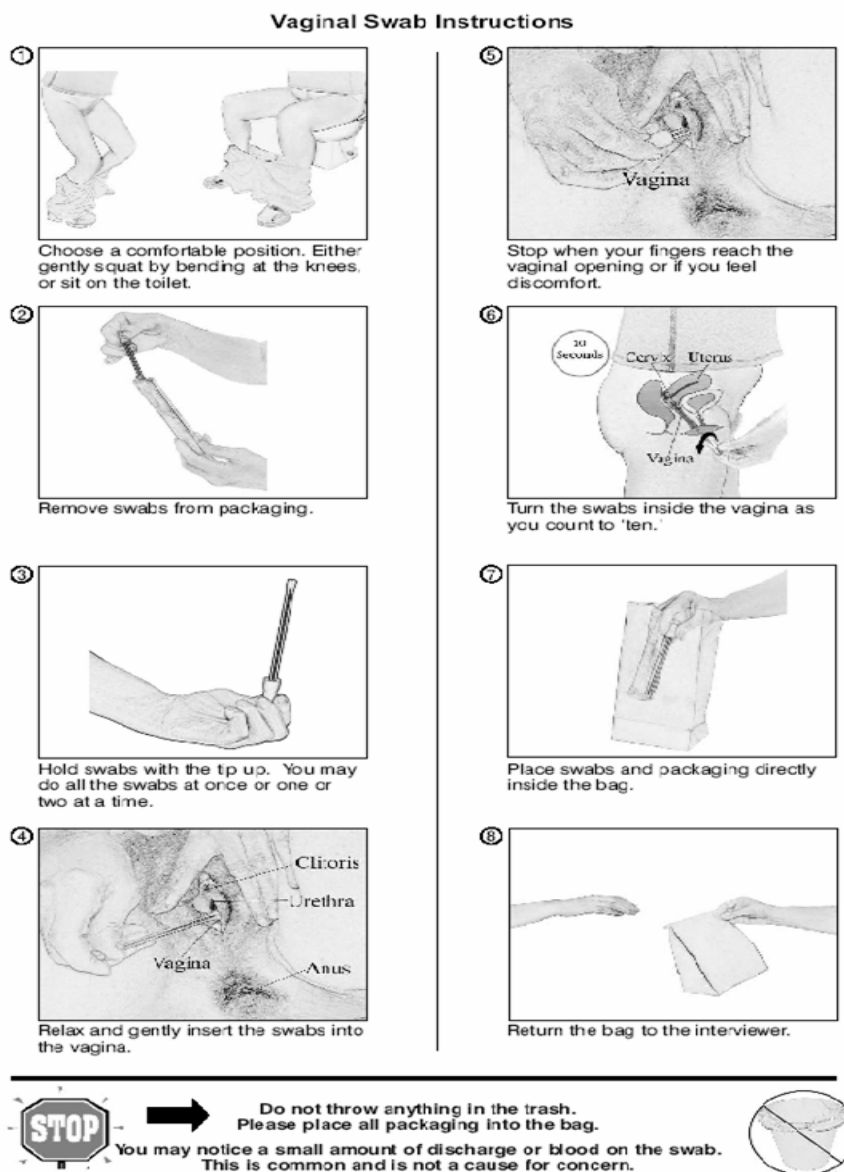
*HPV positivity includes both high-risk and low-risk subtypes

Specimen Collection (Females Only)

The Vaginal Swab Protocol was designed to collect vaginal specimens for Bacterial Vaginosis (BV), Vaginal Candidiasis (VC), Human Papillomavirus (HPV) testing, and vaginal cytology analysis. All female respondents were asked to provide a vaginal self-swab specimen. Procedures were explained using a scripted description aided by illustrated instructions, developed for NSHAP by a medical illustrator in conjunction with a study investigator who is a gynecologist (Figure 1). Participant questions were addressed using a "frequently asked questions" document to ensure consistency of responses across field staff. Field staff read each step of the illustrated instructions to the respondent and asked for questions. Participants were given the instruction card with the collection materials (Female Swab Specimen Collection Kit, Catalog No. 5123-1220; Digene Corporation, Gaithersburg, MD and BBL™ CultureSwab™ Plus, Catalog No. 220117; Becton, Dickinson and Company, Franklin Lakes, NJ) and directed to a

bathroom or other private room in the home. When the respondent returned, the interviewer then secured the Digene swab inside a tube containing 1 mL Specimen Transport Medium™ (STM; Digene Corp.) and the BBL™ CultureSwab™ inside a tube containing Amies medium without charcoal (Becton, Dickinson and Company). The interviewer labeled both tubes with the unique, numeric identification number. At the end of each home encounter, field staff stored the vaginal swab transport tubes in an insulated cooler with ice packs. Vaginal swabs were shipped daily on cold packs in a Styrofoam container to the University of Pittsburgh, Magee-Women's Hospital Department of Pathology clinical microbiology laboratory via overnight delivery. The swabs were packaged in accordance with the federal shipping guidelines for diagnostic biological material. Following processing at Magee-Women's, one BBL™ CultureSwab™ for each respondent was repackaged and shipped overnight on cold packs to the University of Chicago Institute for Mind and Biology laboratory for cytological analysis. An interactive reconciliation system facilitated remote tracking of vaginal swabs. Vaginal swab specimens were collected from all willing female respondents (n = 1,028), with an adjusted cooperation rate of 67.6%

Figure 1. Self Collection of Vaginal Epithelial cells using Dacron Swab



Instructions developed by medical illustrator Rachel Seelen in conjunction with Stacy Lindau, MD.

Shipping and Storage

After collection, the vaginal swab tubes were given to the interviewer and stored at 2- 8°F, in an insulated bag with two reusable ice packs. Specimens were shipped daily.

Prior to shipping, collection tubes containing specimens were removed from the insulated bag, placed in a small Ziploc bag with a handful of cotton balls, and then placed in a Styrofoam™ container and 8" x 7" x 7" cardboard box with a disposable ice pack. They were shipped to The University of Pittsburgh Department of Pathology, Magee-Women's Hospital Clinical Microbiology Laboratory by FedEx Express, by placing the package in a drop box or calling for pickup by FedEx.

<i>Method</i>	FedEx Express: Placed in drop box or picked up by FedEx
<i>Shipping Address</i>	Jeanne Jordan Magee-Women's Hospital Clinical Microbiology Lab 300 Halket Street, Room 4680 Pittsburgh, PA 15213 (412) 641-4104

Assay

Table 2: NSHAP Vaginal Swab assay used by Magee-Women's Research Institute	
Digene Hybrid Capture 2 (hc2) HPV DNA Test	
Assay principle	In <i>vitro</i> nucleic acid hybridization assay with signal amplification using microplate chemiluminescence
Intended use	Qualitative detection of 13 high risk HPV types
Regulatory status	FDA Approved
Analytical Sensitivity	1.09 pg/mL [0.97-1.27 pg/mL]
Cross Reactivity	HPV 13, 6, 42

High-risk HPV Detection by HC2 Assay

High-risk (HR) HPV DNA testing was performed using the HC2 assay method (Digene Corp.) and rapid capture system (RCS) according to the manufacturer's protocol. The HR HC2 signal amplification assay contains a cocktail of probes complementary to 13 HR HPV types including 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59 and 68. In addition to the kit controls, an external sample processing control consisting of approximately 5,000 HeLa cells, which contain HPV 18 DNA was processed with each batched of specimens to control of adequate cell lysis. Specimens with Relative Light Unit/Cutoff Values equal to or greater than 1.0, obtained using the DML 2000 instrument were considered positive for HR-HPV DNA. Both a negative control and a high-risk HPV calibrator are included in the kit. Each control is tested in triplicate within each run. For the run to be valid, the negative control %CV must be at or below 25%, and the calibrator %CV must be at or below 15%.

HPV Genotyping Test

HPV genotyping steps were carried out on all HC2 HR-HPV positive specimens using the Linear Array HPV Genotyping Test and Linear Array Detection Kit (Roche Molecular Systems Inc., Pleasanton, CA) according to the manufacturer's instructions. The Linear Array HPV Genotyping Test is intended for research use only, and not for use in diagnostic procedures. To be compatible with the Qiagen MinElute Spin Column protocol for DNA extraction (Cat. # 57704, Qiagen Inc. Valencia, CA), 250 microliters of each denatured STM was re-acidified to a pH just below 7.0 prior to DNA purification. Fifty microliters of the 120 microliters elution volume were added to an equal volume of 2X PCR master mix containing hot-start *Taq polymerase* and biotin-labeled PGMY09/11 primer set to amplify 37 different anogenital HPV genotypes; They include the following HR-HPV and low-risk types of HPV: 6, 11, 16, 18, 26, 31, 33, 35, 39, 40, 42, 45, 51, 52, 53, 54, 55, 56, 58, 59, 61, 62, 64, 66, 67, 68, 69, 70, 71, 72, 73, 81, 82, 83, 84, IS39 and CP6108. The HPV target is approximately 450 base pairs in length. In the same PCR reaction, a 268 bp target of the human beta-globin genes is co-amplified and served as an internal control for DNA amplification. Thermocycling was performed using the Applied Biosystems Gold-plated GeneAmp[®] 9700 thermocycler. Amplicon denaturation and detection was carried out using the Linear Array Detection Kit according to the manufacturer's directions. External known negative and known positive specimens were included in each run. Two independent readers interpreted the results. Discordant results were re-reviewed by the initial two readers to resolve any discrepancy.

Quality Control

The laboratory participates in semi-annual proficiency testing for High-risk HPV testing using the HC2 assay, and must achieve greater than 90% overall agreement with these challenges.

Reproducibility Study

A total of 20 STM-based specimens (10-positive and 10 negative) were tested in replicates of four on each of five days, for a total of 20 replicates. According to the manufacture's package insert, STM-based specimens with a mean RLU/CO of 20% or more above the cutoff were positive 100% of the time, and those with a mean RLU/CO of 20% or less below the cutoff were negative 100% of the time. Those specimens that were close to the cutoff yielded approximately equal numbers of positive and negative results.

Effect of potentially interfering substances of STM specimens: The effects of 4 agents were studied (blood, douche, anti-fungal cream and contraceptive jelly). None of the 4 agents added to STM specimens resulted in any false positive results. It was noted that a false negative result could occur with specimens containing low levels of high-risk HPV DNA (approximately 1 pg/ml) in the presence of high concentrations of anti-fungal cream or contraceptive jelly.

Analytical Sensitivity

The manufacturer established the analytical sensitivity using plasmid DNA. The mean limit of detection for all 13 types of HPV was 1.08 pg/ml (95% CI= 0.95-1.25, with a SD = 0.05).

Cross Reactivity

According to the manufacturer cross reactivity occurs with HPV type 6, and 42 when present at concentrations above or at 4 ng/ml. Cross reactivity has also been reported for HPV types 40, 53 and 66.

Availability

STM Vaginal Swab		Blue-Tipped Vaginal Swab (double swab)
Product Name	Female Swab Specimen Collection Kit *Collection kit includes: sterile Dacron® swab and tube with 1 mL of Specimen Transport Medium™	BD BBL™ CultureSwab™ Plus Amies without Charcoal
Manufacturer	Digene Corporation	Becton, Dickinson and Company
Location of Manufacturer	Gaithersburg, MD	Franklin Lakes, NJ
Product Number	5123-1220	220117
Interviewer Instructions	When respondent returns, the interviewer inserts the swab into the STM tube and breaks off the extra handle of the swab by pressing it along the side of the tube. The interviewer then labels the tube with the respondent's SUID number using a Sharpie pen and a blank lab label.	When respondent returns, remove cap from tube with gel and tightly insert blue tipped swab in gel. Label tube with Respondent's SUID number using a Sharpie pen and a blank lab label.

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Vaginal Candidiasis

Vaginal Swab Measurement of Candidiasis in Wave I of the National Social Life Health & Aging Project

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Rationale

Vaginal candidiasis, is caused by the overgrowth of a fungal species, *Candida albicans*, in the vaginal flora (Sobel, Faro et al. 1998). The symptoms of vulvovaginal candidiasis include pruritus (itching), soreness, change in vaginal discharge, and dyspareunia (Sobel 1997; Sobel 2007), and can disrupt sexual and social functioning (Nyirjesy, Peyton et al. 2006). Vaginal candidiasis can be sexually transmitted or associated with sexual activity, but commonly occurs in women who are not sexually active

(Lindau, Hoffmann et al. Under Review). Risk factors associated with vaginal candidiasis include elevated estrogen, diabetes mellitus, use of antibiotics and immune suppression (Sobel 2005).

Measurement

Physical examination often reveals discharge that is characterized as thick, adherent, and "cottage cheese-like." However, the discharge may also be thin and loose, and may be similar to the discharge found in other types of vaginitis. The vaginal pH is typically 4 to 4.5, which distinguishes candidiasis from trichomoniasis or bacterial vaginosis (where the pH is elevated) (Sobel 2007). Clinical diagnosis involves microscopic examination of the vaginal secretions to diagnose and characterize vaginal infections.

Population Prevalence

In 20% of healthy asymptomatic women, *Candida* species can be found in the lower genital tract flora (Sobel 2005). Vulvovaginal candidiasis is common in adults: 70-75% of women are affected by this infection at least once in their lives. Nearly 50% of adult women will experience a second case and 5-8% of women will report four or more episodes (Sobel 2007). Among clinical populations of postmenopausal women not using estrogen therapy, vulvovaginal candidiasis is less common than in younger women (Sobel 2005).

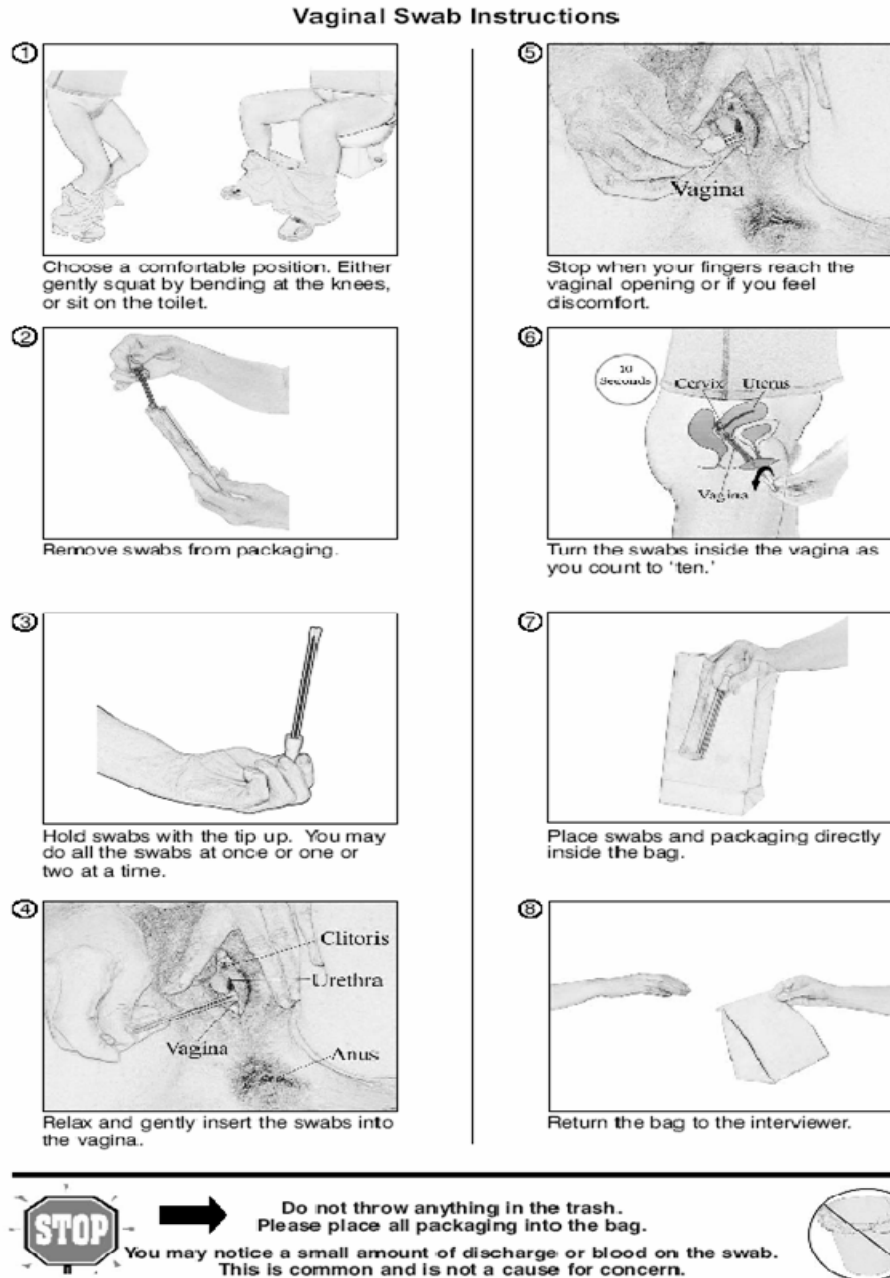
- In a study conducted at Ghent University Hospital over the course of 14 weeks, 612 patients (237 post-menopausal) participated. The participants included both those who had complaints of vaginitis and those who did not. Samples from each participant were taken from the lateral vaginal walls and were identified for yeast colonization. This study showed that yeast was present in 20.1% of the overall sample and 18.6% of the post-menopausal women (Bauters, Dhont et al. 2002).
- In a longitudinal study of 1248 asymptomatic women ages 18-30 years, conducted over a period of four hospital visits in one year, 1,248 asymptomatic (18-30 years old) participated. In this study, yeast was present in 70% of the women at least once and in 30% of the women during all four visits over a one year period (Beigi, Meyn et al. 2004).

Specimen Collection

The Vaginal Swab Protocol was designed to collect vaginal specimens for Bacterial Vaginosis (BV), Vaginal Candidiasis (VC), Human Papillomavirus (HPV) testing, and vaginal cytology analysis. All female respondents were asked to provide a vaginal self-swab specimen. Procedures were explained using a scripted description aided by illustrated instructions, developed for NSHAP by a medical illustrator in conjunction with a study investigator who is a gynecologist (figure 1). Participant questions were addressed using a "frequently asked questions" document to ensure consistency of responses across field staff. Field staff read each step of the illustrated instructions to the respondent and asked for questions. Participants were given the instruction card with the collection materials (Female Swab Specimen Collection Kit, Catalog No. 5123-1220; Digene Corporation, Gaithersburg, MD and BBL™ CultureSwab™ Plus, Catalog No. 220117; Becton, Dickinson and Company, Franklin Lakes, NJ) and directed to a bathroom or other private room in the home. When the respondent returned, the interviewer then secured the Digene swab inside a tube containing 1 mL Specimen Transport Medium™ (STM; Digene Corp.) and the BBL™ CultureSwab™ inside a tube containing Amies medium without charcoal (Becton, Dickinson and Company). The interviewer labeled both tubes with the unique, numeric identification number. At the end of each home encounter, field staff stored the vaginal swab transport tubes in an insulated cooler with ice packs. Vaginal swabs were shipped daily on cold packs in a Styrofoam container to the University of Pittsburgh, Magee-Women's Hospital Department of Pathology clinical microbiology laboratory via overnight delivery. The swabs were packaged in accordance with the federal shipping guidelines for diagnostic biological material. Following processing at Magee-Women's, one BBL™ CultureSwab™ for each respondent was repackaged and shipped overnight on cold packs to the University of Chicago Institute for Mind and Biology laboratory for cytological analysis. An interactive

reconciliation system facilitated remote tracking of vaginal swabs. Vaginal swab specimens were collected from all willing female respondents (n = 1,028), with an adjusted cooperation rate of 67.6% (Lindau, Hoffmann et al. Under Review).

Figure 1. Self Collection of Vaginal Epithelial cells using Dacron Swab



Instructions developed by medical illustrator Rachel Seelen in conjunction with Stacy Lindau, MD.

Shipping and Storage

After collection, the vaginal swab tubes were given to the interviewer and stored at 2-8°F, in an insulated bag with two reusable ice packs.

Prior to shipping, collection tubes containing specimens were removed from the insulated bag, placed in a small Ziploc™ bag with a handful of cotton balls, and then placed in a Styrofoam™ container and 8" x 7" x 7" cardboard box with a disposable ice pack. They were shipped to The University of Pittsburgh Department of Pathology, Magee-Women's Research Institute Laboratory by overnight FedEx Express. The swabs were packaged in accordance with the federal shipping guidelines for diagnostic biological material and were placed in a drop box or calling for pickup by FedEx. Specimens were shipped daily.

Weekly reconciliation, using a remote and interactive vaginal swab tracking system, between NORC and the Magee-Women's laboratory, compared questionnaire data (indicating whether a sample was recorded as collected) with laboratory data (indicating whether a sample was recorded as received). Discrepancies in reconciliation were resolved and recorded. (Lindau, Hoffmann et al. Under Review)

<i>Method</i>	FedEx Express: placed in drop box or picked up by FedEx
<i>Shipping Address</i>	Jeanne Jordan Magee-Women's Hospital Clinical Microbiology Lab 300 Halket Street, Room 4680 Pittsburgh, PA 15213 (412) 641-4104

Assay

Preparation for the vaginal Candidiasis assay began with rolling of the culture swab onto the surface of a glass slide, thus transferring the vaginal cellular specimen from the swab to the slide. The slide was air dried and fixed with 95% methanol and then stained using a standard gram stain protocol. (Burke 1922)

The same gram stained slides used to assess the presence of bacterial vaginosis, were also evaluated for the presence of yeast, by using the 1000x objective under oil. A dichotomous score was assigned to each specimen slide, after the reader viewed and examined the entire area of each stained slide to determine the presence or absence of yeast cells showing blastoconidia (cell buds) (Lindau, Hoffmann et al. Under Review).

Technical details of the vaginal swab laboratory assays have been previously described (Lindau, Drum et al. Under Review). The vaginal Candidiasis assay method is detailed below:

Table 1. Summary of Vaginal Swab Assays Obtained from Self-Swab Specimens in NSHAP^f					
Assay Type	Collection Device	Manufacturer	Laboratory	Assay Type	Protocol
Vaginal Candidiasis (VC)	Double Copan swab with Dacron tip	BBL™ CultureSwab™ Plus, Catalog No. 220117; Becton, Dickinson and Company, Franklin Lakes, NJ	UP-MWRI	Gram stain of vaginal material on glass slide (Burke, 1922)	Clinical

Magee-Womens Research Institute laboratory performed a Gram stained slide of a smear made from the vaginal swab samples. The smear was read for the presence of yeast, which stains gram-positive.

Table 2. NSHAP Vaginal Swab testing performed by Magee-Women's Research Institute			
Assay Type	Assay principle	Regulatory Status	Samples
Gram stain smear	Evaluation for the presence of yeast	None	Vaginal or cervical self-collected swab

Standard clinical procedures were used by Jeanne A. Jordan, Ph.D., Associate Professor in Pathology, University of Pittsburgh, Director-Microbiology, Virology, Molecular Diagnostics and Immunology at Magee-Women's Hospital, Associate Director, Magee-Women's Research Institute (Burke et al.).

Performance Characteristics

Each stained slide was read by two independent observers who were blinded to the other's results. Discordant results were re-reviewed by the initial two readers as well as reviewed by a third observer to resolve the discrepancies.

Quality Control

The laboratory participates in semi-annual proficiency testing for gram staining, and must achieve greater than 90% overall agreement with these challenges. Gram stain reagents are checked weekly with each new lot of stain before it is put into use. Gram stain reagents were evaluated by staining ATCC 25923 *Staphylococcus aureus*, ATCC 25922 *E. coli*, and ATCC 26555 *Candida albicans*.

Availability

STM Vaginal Swab		Blue-Tipped Vaginal Swab (double swab)
Product Name	Female Swab Specimen Collection Kit *Collection kit includes: sterile Dacron® swab and tube with 1 mL of Specimen Transport Medium™	BD BBL™ CultureSwab™ Plus Amies without Charcoal
Manufacturer	Digene Corporation	Becton, Dickinson and Company
Location of Manufacturer	Gaithersburg, MD	Franklin Lakes, NJ
Product Number	5123-1220	220117
Interviewer Instructions	When respondent returns, insert the swab into the STM tube and break off the extra handle of the swab by pressing it along the side of the tube. Label the tube with Respondent's SUID number using a Sharpie pen and a blank lab label.	When respondent returns, remove cap from tube with gel and tightly insert blue tipped swab in gel. Label tube with Respondent's SUID number using a Sharpie pen and a blank lab label.

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